



Joe Snacks Infrastructure Design Project

NET 377

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To: Joseph Jasper

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1. Overview

1.1: About Me:

I am Sukaina Hussain, majoring in Cybersecurity at DePaul University. I am also a Network security analyst and assigned to propose Joe Snacks (JS) a network design. I have experienced the best enterprise security practices and layered security models. Also, my expertise includes security system rule design, Static routing, NAT implementation and Secure Server placement, which is all very much essential in constructing your network design.

1.2: Joe Snacks Network Components:

1.2.1: Servers

- JS-Sales Server (HTTPS – Public)
- DB- Server (PostgreSQL – internal database)
- HR-Server (HTTPS – internal employees)
- RD-Gateway (Remote Desktop Protocol (RDP) for remote employees)
- AD-Server (LDAP authentication)
- WSA-Gateway (Web proxy for employee web browsing)

1.2.2: Network Devices and Subnets

- 2 ISP routers
- 4 ASA Firewalls
- 3 Layer 2 switches
- 2 Layer 3 switches
- 6 Servers
- Subnets:
 - R-1 Subnet
 - DMZ Subnet
 - Internal Subnet
 - Intranet Subnet
 - G1-Subnet

1.3: Security Challenges:

Joe Snacks (JS) only have two public IP addresses provided by ISP which requires the use of Network Address Translation (NAT) from Public Network (Internet) to Private Network (JS

Internal Network). NAT allows multiple devices on a private local network to share a single, public IP address to access the internet. However, NAT also increases the complexity of translating internal private addresses to public IP addresses and ensures correct routing and firewall rules for translated traffic.

Secure Remote employee access is also a challenge because it is frequently targeted by attackers, it is not exposed to public but still prone to internal threats. The network must ensure authentication through Active Directory and implementing Group policies on the Employee users to access what they need to access only which is called Principle of least privilege (PoLP).

2. Network Architecture

2.1: Dual-Path Dual-Layer Firewall Architecture

The Dual-Path Dual-Layer firewall architecture uses two separate traffic paths and two firewall layers to protect the internal network. Red Rail manages the incoming traffic, and Green Rail manages the outgoing traffic. Each Firewall only allows new connections in one direction, which simplifies the security policy enforcement and improves network protection.

To understand why this is the most suitable architecture for Joe Snacks, I will compare Dual-Path Dual-layer firewall architecture with other available architecture.

2.2: Two-Leg Firewall Architecture:

A two-leg firewall architecture connects a single firewall between the Internet and the internal network. All inbound and outbound traffic must pass through that single firewall device.

However, relying on a single firewall introduces significant security limitations. Studies on firewall deployment emphasize that a single perimeter firewall is insufficient for modern enterprise environments because it creates a single point of failure and lacks layered protection.

Other limitations include, we cannot implement DMZ to isolate public servers from untrusted external network (internet), If the Firewall is compromised, the internal systems

will directly be exposed to public network and there is limited segmentation between services accessed by customers and employees.

For the JS network, this architecture would be unsafe because the JS-Sales-Server must be publicly accessible, while critical servers such as AD-Server and DB-Server must remain protected inside the network.

2.3: Three-Leg Firewall Architecture:

A three-leg firewall architecture introduces a DMZ network by adding a third interface on the firewall. This allows Internet-facing servers to be placed in the DMZ while internal resources remain separated.

Although this design improves security compared to the two-leg architecture, it still relies on a single firewall device for enforcing all security policies.

Three-leg does improve segmentation of network but there is still single point of failure risk and more complex rulesets because firewall manages traffic in both directions.

2.4: Split Firewall Architecture:

A split firewall architecture improves security by deploying two firewalls instead of one. In this design, one firewall sits between the Internet and the DMZ. Another firewall sits between the DMZ and the internal network. This design implements layered protection, which aligns with defense-in-depth strategies.

However, both inbound and outbound traffic still pass through the same network path, which can create complex firewall rule sets and increase the risk of configuration errors.

2.5: Analysis of Architectures:

From the above discussed the benefits and risks of implementation for the JS network design, dual-path dual-layer Firewall architecture is best and suits the security of JS network critical servers.

Firstly, this architecture uses multiple firewall layers, ensuring that if one firewall is compromised, the second layer still protects the internal network. Multi-layer security designs have been shown to significantly improve network resilience against attacks.

Secondly, the Traffic direction separation allows to impose more security on critical servers such as HR-server, which can be only accessible by the internal employees, and not exposed to public. It also enforces simpler rules that allow new connections in one direction which reduces configuration errors.

Thirdly, there is strong and clear Network segmentation, which is important for protecting sensitive resources by isolating systems with different security requirements. The architecture separates Internet Users, DMZ servers, Internal Servers, and employee workstations.

In conclusion, implementing Dual-Path Dual-layer architecture, improves the overall security of the network as it reduces the attack surface with the help of segmentation and Defense-in-depth. It is well suited for JS infrastructure because it supports both public services and protects internal systems.

2.6: Joe Snacks Network diagram:

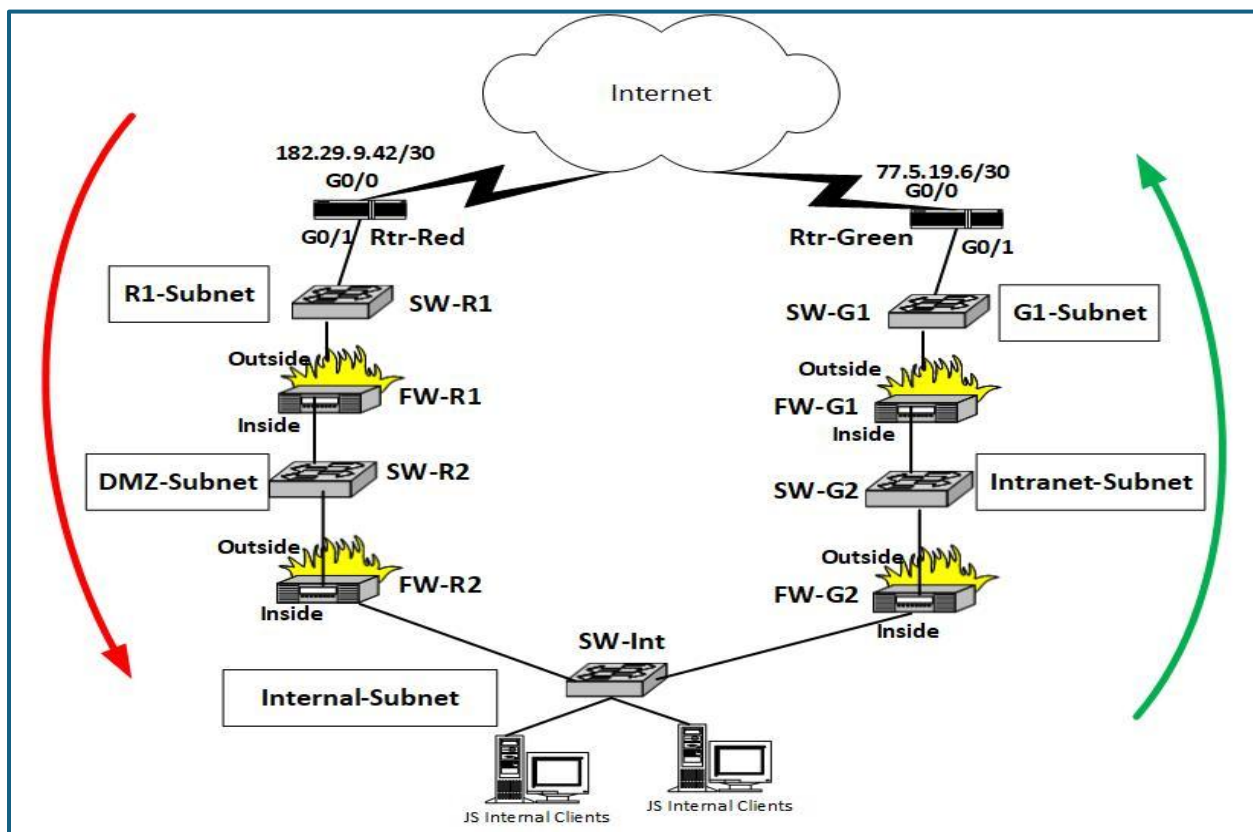


Fig 2.6: Dual-layer Dual Path Network diagram

3. IP Addressing

JS Network IP Subnets					
Subnet Name	Network Address	Prefix Length (/n)	Number of Assignable IPs	First Assignable IP	Last Assignable IP
R1-Subnet	10.19.2.128 /29	/29	6	10.19.2.129	10.19.2.134
DMZ-Subnet	10.19.2.0 /26	/26	62	10.19.2.1	10.19.2.62
Internal-Subnet	10.19.0.0 /23	/23	510	10.19.0.1	10.19.1.254
Intranet-Subnet	10.19.2.64 /26	/26	62	10.19.2.65	10.19.2.126
G1-Subnet	10.19.2.136 /29	/29	6	10.19.2.137	10.19.2.142

Fig 3.1: IP addressing for subnets.

In Fig 3.1, I have assigned IP blocks to different subnets according to the network requirements.

For internal subnet, this was the largest subnet block which requires two hundred wired workstations, 20 Access points, switch management, Firewall interfaces, and prospects of future growth of JS network. I assigned /23 block, which gave 510 assignable IP addresses.

For DMZ subnet and intranet subnet separately, the requirements were up to forty servers, firewall interface and switch management which needed Ips for estimated forty-five devices. I assigned /26 block which gives sixty-two assignable IP addresses, keeping in mind the future expansion plans.

For R1 subnet and G1 subnet separately, they had only point-to-point connections between router and firewall interface and needed another IP for Switch VLAN interface thus I chose a block of /30, which gives six assignable Ips for each subnet.

JS Network Device Interface IPs				
Device Name	Interface	Subnet Name	IP Address	Subnet Mask
Rtr-Red	G0/0	ISP-Link1	182.29.9.42	255.255.255.252
Rtr-Red	G0/1	R1-Subnet	10.19.2.129	255.255.255.248
Rtr-Green	G0/0	ISP-Link2	77.5.19.6	255.255.255.252
Rtr-Green	G0/1	G1-Subnet	10.19.2.137	255.255.255.248
FW-R1	Inside	DMZ-Subnet	10.19.2.1	255.255.255.192
FW-R1	Outside	R1-Subnet	10.19.2.130	255.255.255.252
FW-R2	Inside	Internal-Subnet	10.19.0.1	255.255.254.0
FW-R2	Outside	DMZ-Subnet	10.19.2.2	255.255.255.192
FW-G2	Inside	Internal-Subnet	10.19.0.2	255.255.254.0
FW-G2	Outside	Intranet-Subnet	10.19.2.65	255.255.255.192
FW-G1	Inside	Intranet-Subnet	10.19.2.66	255.255.255.192
FW-G1	Outside	G1-Subnet	10.19.2.138	255.255.255.252
SW-R1	VLAN1	R1-Subnet	10.19.2.131	255.255.255.252
SW-R2	VLAN1	DMZ-Subnet	10.19.2.3	255.255.255.192
SW-Int	VLAN1	Internal-Subnet	10.19.0.3	255.255.254.0
SW-G2	VLAN1	Intranet-Subnet	10.19.2.67	255.255.254.0
SW-G1	VLAN1	G1-Subnet	10.19.2.139	255.255.255.248

Fig 3.2: Interface IP addresses for every device

4. Server Placement and Addressing

In JS network design, servers are placed in different network segments based on security requirements, access needs, and communication direction through red and green rails. The network uses two protected server segments: the DMZ Subnet and the intranet subnet.

The DMZ subnet is designed to host servers that must communicate with external internet users. The servers are placed in DMZ because it should be isolated from internal networks so that if one of them is compromised, attackers cannot have direct access to sensitive internal resources.

The intranet subnet hosts internal servers that store sensitive information about employees, company and provide internal authentication. These systems should only be accessible to internal employees, which is why it is placed on green rail as no public device can access it.

JS-Sales server is placed in DMZ subnet because it hosts company's e-commerce file which should be accessible by customers on internet. This server must accept the in-

bound traffic on the from the internet through Red Rail, it must be in the DMZ where it is isolated from internal systems.

RD-Gateway Server is also placed in the DMZ Subnet because it must accept Remote Desktop connections from employees working remotely. External users connect to this server from the Internet before gaining access to their internal workstations. Since the RD-Gateway must receive inbound connections through the Red Rail, it must be positioned in the DMZ network.

DB-Server is in Intranet Subnet because it stores sensitive sales and order information for the company. The database does not need to be directly accessible from the Internet. Thus, it is placed in the green rail so no traffic can come from internet to DB-Server.

Active Directory Server is in Intranet Subnet because it provides authentication services for internal users and systems. Active Directory contains sensitive identity and credential information that must be protected from external access.

WSA (Web Security Appliance) Server is placed in the Intranet Subnet because it acts as a secure web proxy for internal employees. The WSA server inspects and filters web traffic before allowing it to reach external websites, helping protect the network from malware, malicious websites, and other Internet-based threats. Because the WSA server primarily serves internal users and initiates outbound connections to the Internet through Green Rail, it is best located in the intranet subnet.

JS Network Server Placement and IPs				
Server Name	Subnet Name	Server IP	Subnet Mask	Default Gateway IP
JS-Sales-Server	DMZ-Subnet	10.19.2.4	255.255.255.192	10.19.2.3
DB-Server	Intranet-Subnet	10.19.2.68	255.255.255.192	10.19.2.67
HR-Server	Intranet-Subnet	10.19.2.69	255.255.255.192	10.19.2.67
RD-Gateway	DMZ-Subnet	10.19.2.5	255.255.255.192	10.19.2.3
AD-Server	Intranet-Subnet	10.19.2.70	255.255.255.192	10.19.2.67
WSA-Gateway	Intranet-Subnet	10.19.2.71	255.255.255.192	10.19.2.67

Fig 4.1: JS server placements and IP addresses.

5. Network Address Translations (NAT)

NAT allows private internal IP addresses to communicate with external networks (such as the Internet) by translating them into a public IP address. Since private IP addresses are not routable on Public Internet, NAT allows internal devices to excess external devices while

hiding their actual internal address. Thus, NAT also improves security by preventing external systems from directly seeing or connecting to internal private addresses.

In the JS network, NAT is required because all internal devices use private addresses from the 10.19.0.0/16 network block, while the Internet uses public IP addressing. Additionally, the company only has two public IP addresses provided by the ISP, which must be shared among multiple internal systems. NAT allows the network to map internal servers and devices to these limited public IP addresses while maintaining secure connectivity with external users and services.

JS Network NAT Rule Specifications										
Inside Interface	Outside Interface	NAT(N) or PAT (P)	Static (S) or Dynamic (D)	Inside IP (First)	Inside IP (Last or Mask)	Outside IP or Interface	Protocol (if specified)	Inside Port Number (if needed)	Outside Port Number (if needed)	Comments
g0/0/1	g0/0/0	N	S	10.19.2.4	host	182.29.9.42	tcp	443	443	JS-Sales-Server
g0/0/1	g0/0/0	N	S	10.19.2.5	host	182.29.9.42	tcp	443	3389	RD-Gateway
Rtr-Green	g0/0/1	g0/0/0	P	10.19.2.64	0.0.0.63	g0/0/0				WSA-gateway

Fig 5.1: shows the NAT configuration on Routers.

5.1: NAT Operations

The following scenarios demonstrate how NAT is used in the JS network to control traffic between external users, DMZ servers, and internal resources.

1. An Internet user at public IP **4.5.6.7** connects to JS-Sales-Server to order products.

The internet requests public IP address 4.5.6.7 attempts to access JS-Sales Server to place an order. The external user sends a request to the public IP address 182.29.9.42 on TCP port 443, which is the public address assigned to the JS-Sales web service. When the request reaches the Red Router (Rtr-Red), the router checks its NAT table and identifies a static NAT rule that maps the public address 182.29.9.42:443 to the internal private IP address 10.19.2.4, which belongs to the JS-Sales Server located in the DMZ subnet.

The router then performs destination NAT, replacing the destination address in the packet from 182.29.9.42 to 10.19.2.4, and forwards the packet to the DMZ network through interface g0/0/1. The JS-Sales Server processes the request and sends a response back to the user. The router performs the reverse NAT translation so that the response appears to originate from the public IP 182.29.9.42.

This configuration allows customers on the Internet to securely access the sales application using HTTPS, while keeping the actual internal IP address of the server hidden.

2. WSA-Gateway sets up a new HTTP connection to public webserver at IP address 22.3.4.5

WSA Gateway server initiates an outbound HTTP request to an external web server located at 22.3.4.5.

The WSA Gateway is located inside the internal network and uses the private address 10.19.2.71. When the gateway sends a packet toward the Internet, the packet reaches the Green Router (Rtr-Green). The router applies to the configured dynamic NAT with Port Address Translation (PAT) rule for the internal network 10.19.2.0/26.

The router replaces the source private address 10.19.2.71 with the public IP address assigned to its external interface g0/0/0, while also assigning a unique port number. The translated packet is then sent to the external web server at 22.3.4.5.

When the web server responds, the packet returns to the router's public IP address and port. The router consults its NAT translation table and maps the packet back to the original internal address 10.19.2.71, forwarding the response to the WSA Gateway.

3. Internal Client contacts HR-Server to check on their vacation time.

In this scenario, an internal client connects to the HR Server to check vacation information.

Both the client device and the HR Server are located within the internal intranet subnet. Because communication occurs entirely within the private network, the packets are routed internally without requiring NAT translation.

The client sends a request directly to the private IP address of the HR Server, and the network router simply forwards the packet based on its internal routing table. The HR Server processes the request and returns the response directly to the client.

6. Static Routes:

The JS network uses **static routing** to control traffic flow between the internal intranet network, the DMZ subnet, and the external Internet connection. Static routes are manually configured on the routers to ensure predictable and secure routing behavior. Because the network topology is small and stable, static routing provides a simple and efficient solution without the overhead of dynamic routing protocols.

Static routing also improves security by allowing administrators to explicitly define the paths that traffic can take through the network. This reduces the risk of routing

misconfigurations or malicious route advertisements that could occur with dynamic routing protocols.

6.1: Router Red

The Red Router acts as the external gateway that connects the network to the Internet and the DMZ segment. This router must know how to reach the DMZ subnet thus it has two static routes; one is the default route to internet and the other is route to DMZ subnet. Red router does not have static route to intranet subnet as the internet users only need to access the DMZ servers, they will not have access to the intranet servers. This provides another layer of security to protect the servers from getting compromised.

6.2: Green Router

The Green Router is an edge router between the internal intranet subnet and the Internet. Because the Green Router is responsible for forwarding all external traffic, it requires a default route pointing toward the internet to send the traffic back to the internet. It also has a static route to the intranet subnet because it to route the returning traffic from intranet.

6.3: Firewall-R1

Firewall-R1 is sitting between R1 subnet and DMZ subnet, it only has a default route to the internet, to forward traffic back to the internet.

6.4: Layer 3 Switch (DMZ)

L3 Switch is sitting between FW-R1 and FW-R2. I chose an L3 switch because I wanted to use it for routing the traffic to internal subnet and intranet subnet. This design choice improves network performance, scalability, and security by separating routing functions from security inspection functions. It helps to reduce load on Firewalls to process the traffic and improve the network performance significantly.

It has one static route to the internal subnet and the other to intranet subnet, as an employee who connects through RD Gateway server can easily route to the internal or intranet subnet through the switch. The default route is to the internet as if anyone from the internal subnet wants to connect to the internet.

6.5: Firewall -R2

This device sits between DMZ subnet and internal subnet. It only has static routes to intranet subnets to forward packets from DMZ servers to intranet servers.

6.6: Firewall-G2

This device is sitting between the internal subnet and intranet subnet; it only has one static route to DMZ to allow access from intranet to the DMZ subnet.

6.7: Layer 3 Switch (Intranet)

This device is sitting between FW-G2 and FW-G1, to make the routing from intranet subnet to internal and DMZ subnets easier. It has total three static routes, two of which are too internal subnet and DMZ subnet, respectively. The last route is default route to the internet to send the traffic from intranet subnet to the internet.

6.8: Firewall-G1

This device is between Layer 3 Switch G2 and Switch G1. It contains the static route to internal subnet, as internet users can access the internal subnet of JS, but the firewall ACLS will determine what can be accessed and from what source. It also contains a default route to internet, to forward traffic back to internet.

JS Network - Static Route Specifications					
Device Name	Destination Network	Destination Mask	Next Hop IP	Outgoing Interface	Comment
Rtr-Red	10.19.2.0	255.255.255.192	10.19.2.130	g0/0/1	to DMZ
	0.0.0.0	0.0.0.0	182.29.9.41	g0/0/0	to internet
Rtr-Green	10.19.2.64	255.255.255.192	10.19.2.138	g0/0/1	to intranet
	0.0.0.0	0.0.0.0	77.5.19.5	g0/0/0	to internet
FW-R1	0.0.0.0	0.0.0.0	10.19.2.129	g0/0/0	to internet
FW-R2	10.19.2.64	255.255.255.192	10.19.0.2	g0/0/1	to intranet
FW-G2	10.19.2.0	255.255.255.192	10.19.0.1	g0/0/1	to DMZ
FW-G1	10.19.0.0	255.255.254.0	10.19.2.67	g0/0/1	to internal subnet
	0.0.0.0	0.0.0.0	10.19.2.137	g0/0/0	to internet
SW-DMZ	10.19.0.0	255.255.254.0	10.19.2.2	vlan 1	to internal subnet
	10.19.2.64	255.255.255.192	10.19.2.2	vlan 1	to intranet subnet
	0.0.0.0	0.0.0.0	10.19.2.1	vlan 1	to internet
SW-Intranet	10.19.2.0	255.255.255.192	10.19.2.65	vlan 1	to DMZ
	10.19.0.0	255.255.252.0	10.19.2.65	vlan 1	to internal subnet
	0.0.0.0	0.0.0.0	10.19.2.66	vlan 1	to internet

Fig 6.1: Static Routes configured on each device

7. Firewall Rules:

The JS network is designed so that incoming and outgoing traffic does not use the same path. New connections that begin with the public Internet enter only on the **Red Rail**, while new connections that begin from inside the company network leave only on the **Green Rail**. This separation simplifies the firewall policy because each firewall only needs to allow new connections in one intended direction. The interface facing the “wrong” direction never needs to allow new sessions, so its ruleset can simply be denied Ip any any. This reduces complexity, makes the rule base easier to audit, and lowers the chance of accidentally permitting traffic that the business does not require. Since the design requirements list a small set of approved TCP connections and explicitly state that all other connections must be denied, the one-way permit model fits this architecture very well.

7.2: Firewalls Rulesets

JS Network Firewall Ruleset Specifications										
(Ruleset Name = "<device>-<interface>" for Incoming packets)										
RuleSet Name	Rule #	P/D	Protocol	Source IP (First)	Source IP (Last or Mask)	Source Port	Dest IP (First)	Dest IP (Last or Mask)	Dest Port	Comments
FW-R1-Inside	1	D	ip	any			any			Only incoming connections allowed on Red Rail
FW-R1-Outside	1	P	tcp	any	any	any	10.19.2.4	host	443	Internet to JS Sales HTTPS
	2	P	tcp	any	any	any	10.19.2.5	host	3389	Internet to RD Gateway
	3	D	ip	any	any	any	any	any	any	Deny all
FW-R2-Inside	1	D	ip	any			any			Only incoming connections allowed on Red Rail
FW-R2-Outside	1	P	tcp	10.19.2.4	host	any	10.19.2.68	host	5432	JS Sales to DB
	2	P	tcp	10.19.2.5	host	any	10.19.2.70	host	389	RD Gateway to AD
	3	P	tcp	10.19.2.5	host	any	10.19.0.0	0.0.1.255	3389	RD Gateway to Internal Clients
	4	D	ip	any	any	any	any	any	any	Deny all
FW-G2-Inside	1	P	tcp	10.19.0.0	0.0.1.255	any	10.19.2.68	host	5432	Internal to DB
	2	P	tcp	10.19.0.0	0.0.1.255	any	10.19.2.70	host	389	Internal to AD
	3	P	tcp	10.19.0.0	0.0.1.255	any	10.19.2.69	host	443	Internal to HR
	4	P	tcp	10.19.0.0	0.0.1.255	any	10.19.2.71	host	80	Internal to WSA
	5	P	tcp	10.19.0.0	0.0.1.255	any	10.19.2.71	host	443	Internal to WSA
	6	D	ip	any	any	any	any	any	any	Deny all
FW-G2-Outside	1	D	ip	any			any			Only outgoing connections allowed on Green Rail
FW-G1-Inside	1	P	tcp	10.19.2.71	host	any	any	any	80	WSA to Internet HTTP
	2	P	tcp	10.19.2.71	host	any	any	any	443	WSA to Internet HTTPS
	3	D	ip	any	any	any	any	any	any	deny all
FW-G1-Outside	1	D	ip	any			any			Only outgoing connections allowed on Green Rail

Fig 7.1: Firewall Rulesets

7.3: Firewall Rulesets Description

1. Internet users to JS-Sales-Server:

The firewall rules must allow public Internet clients to open new HTTPS TCP connections to the JS-Sales-Server in the DMZ so customers can place online orders.

2. JS-Sales-Server to DB-Server:

The firewall rules must allow the JS-Sales-Server to open new PostgreSQL TCP connections to the DB-Server so completed web orders can be written into the database.

3. Internal clients to DB-Server:

The firewall rules must allow internal employee workstations to open new PostgreSQL TCP connections to the DB-Server so sales employees can enter phone orders into the database.

4. Internal clients to AD-Server:

The firewall rules must allow internal employee workstations to open new LDAP TCP connections to the AD-Server so employees can authenticate their logins.

5. Internet users to RD-Gateway:

The firewall rules must allow authorized remote users on the Internet to open new RDP TCP connections to the RD-Gateway so they can begin a remote desktop session securely through the gateway.

6. RD-Gateway to AD-Server:

The firewall rules must allow the RD-Gateway to open new LDAP TCP connections to the AD-Server so remote users can be authenticated before being granted access.

7. RD-Gateway to internal workstations:

The firewall rules must allow the RD-Gateway to open new RDP TCP connections to approved internal employee workstations so authenticated remote users can reach their office computers.

8. Internal clients to HR-Server:

The firewall rules must allow internal employee workstations to open new HTTPS TCP connections to the HR-Server so employees can access employment and benefits information securely.

9. Internal clients to WSA-Gateway:

The firewall rules must allow internal employee workstations to open new HTTP and HTTPS TCP connections to the WSA-Gateway so web requests can be inspected by the security proxy before leaving the network.

10. WSA-Gateway to public web servers:

The firewall rules must allow the WSA-Gateway to open new HTTP and HTTPS TCP connections to public Internet web servers so employee web browsing can occur only through proxy.

8. High Availability Service for Joe Snacks Infrastructure

High Availability is especially important for the red rail in the JS network because this path carries all new inbound connections from Internet users to the JS-Sales-Server. The JS-Sales-Server hosts the company's online ordering system, which means that any outage along this path would immediately prevent customers from placing orders. If a single firewall, switch, or network link on the red rail was to fail, Internet traffic would no longer be able to reach the server in the DMZ, resulting in lost sales, reduced customer trust, and potential monetary loss for the company.

To make the JS network available for Internet users placing orders on the JS-Sales-Server, JS Corporation should add redundancy at three points in the path: switches, firewalls, and links. First, the single edge switch and the single DMZ switch should each be replaced with a pair of switches connected by redundant links. These switch pairs should run Rapid Spanning Tree Protocol or Spanning Tree Protocol so that only one Layer 2 path forwards at a time while the alternate path remains ready if a link or switch fails. This prevents switching loops while still preserving an immediate backup path. Second, the firewall on the inbound path should be replaced by a two-node firewall cluster. The firewall pair should present a shared virtual IP address to the surrounding network, so Internet traffic always targets the same address even if the active firewall fails. Third, each firewall should be dual homed to both switches on each side, so no single cable failure disconnects the service path. With this design, a single switch failure, a single firewall failure, or a single link failure would not interrupt Internet access to the JS-Sales-Server.

The role of virtual IP services is important because they hide the failover process from end systems. Cisco's first-hop redundancy documentation explains that devices can share one virtual IP address, so hosts continue using the same gateway while traffic shifts between members during failover. In a firewall cluster, the same idea applies to the firewall-facing addresses: upstream and downstream devices continue forwarding to the cluster's stable address instead of being reconfigured during an outage. This is what allows failover to happen quickly and transparently.

For firewall redundancy, JS Corporation could choose either an Active/Active cluster or an Active/Passive cluster. Active/Active uses both firewalls to process traffic at the same time, which can improve capacity usage and may provide faster handling of peak traffic.

However, vendor documentation also shows that Active/Active is more complex because traffic distribution and session handling become harder to design correctly. By contrast, Active/Passive is simpler: one firewall actively processes traffic while the standby firewall continuously synchronizes configuration and session state, then takes over if the active unit fails. This model is easier to implement, easier to troubleshoot, and usually better suited for a small environment like JS Corporation, where the main goal is service continuity rather than maximum throughput.

Based on these tradeoffs, I recommend that JS Corporation deploy an Active/Passive firewall cluster on the Internet-to-DMZ path that protects the JS-Sales-Server. The main advantage is operational simplicity: failover behavior is straightforward, state synchronization is easier to validate, and troubleshooting during an outage is less complicated than with Active/Active designs. Since the assignment requirement is to keep ordering available after any single switch, firewall, or link failure, Active/Passive provides the strongest balance of resiliency, cost control, and manageability. Active/Active would make sense mainly if JS Corporation had a larger traffic volume and a stronger need to use both firewalls for performance all the time. For this network, Active/Passive is the safer and more practical high-availability design.

8.1: High-Availability Infrastructure Design

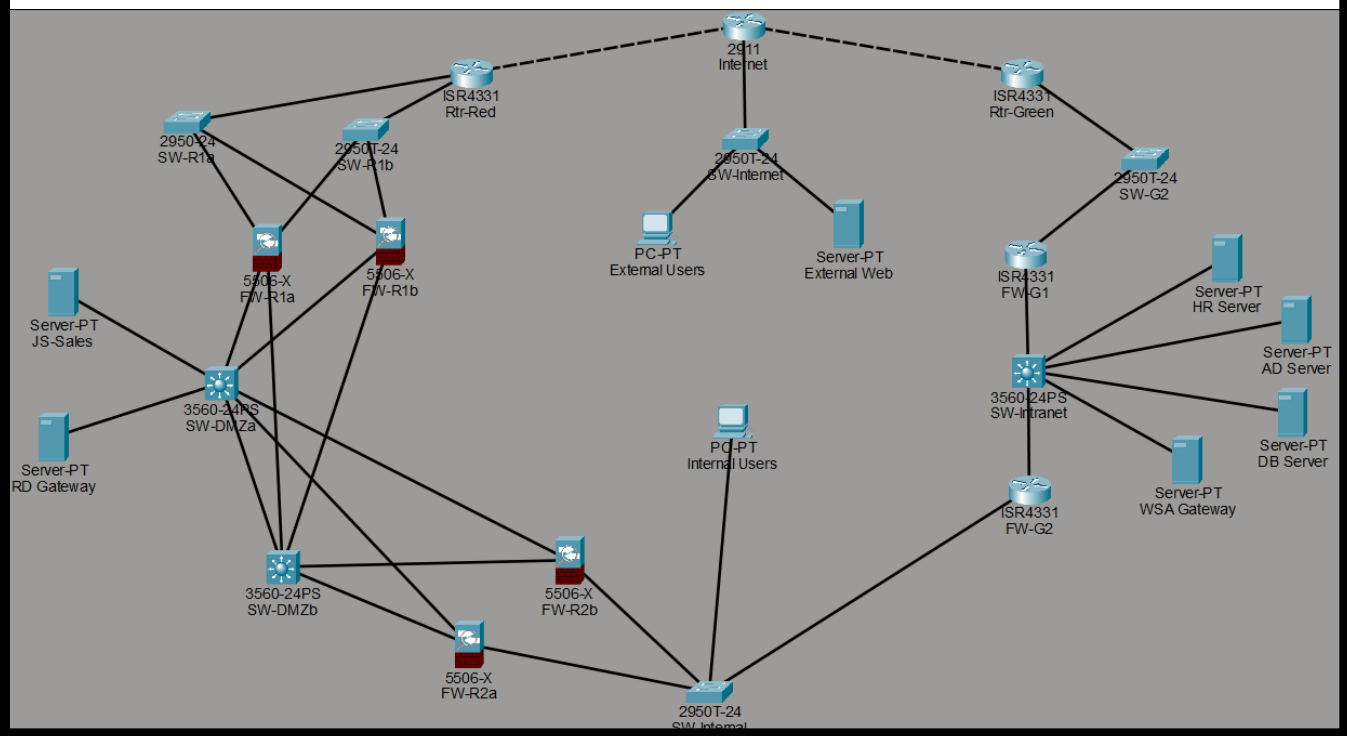


Fig 8.1: High-Availability Infrastructure Design

This infrastructure will provide high availability for Internet users accessing the JS-Sales-Server, the red rail of the network was redesigned to eliminate single points of failure by introducing redundant switches, firewall clusters, and redundant network links. Two switches on the R1 subnet, SW-R1a and SW-R1b, connect the Rtr-Red router to both firewalls FW-R1a and FW-R1b, forming the first firewall cluster. Both firewalls are then connected to two DMZ switches, SW-DMZa and SW-DMZb, which are also interconnected and run Spanning Tree Protocol (STP) to prevent Layer-2 loops while maintaining backup paths. The second firewall layer is also implemented as a cluster using FW-R2a and FW-R2b, which connect to both DMZ switches and the internal network switch SW-Internal. This design ensures that Internet traffic can reach the JS-Sales-Server through multiple paths, so if any single firewall, switch, or network fails, traffic can automatically reroute through an alternate path, and the online ordering service will remain available.

9. Conclusion

In summary, the suggested JS network architecture offers a scalable, secure, and resilient solution that safeguards vital business services while guaranteeing constant availability for clients and staff. The network maintains strong security segmentation between the Internet, DMZ, and internal networks while strictly controlling incoming and outgoing traffic using a dual-path, dual-layer firewall design. By adding high-availability features like redundant switches, firewall clustering, redundant links, and Spanning Tree Protocol, single points of failure are eliminated, and Internet users can continue placing orders on the JS-Sales-Server even if a network device or link malfunctions. Both security and dependability are given top priority in these design choices, which are crucial for preserving client confidence and safeguarding corporate operations.

I approach security analysis with an emphasis on comprehending business needs, identifying potential risks, and putting into practice workable solutions that strike a balance between security, performance, and reliability in addition to putting forth a solid technical design. I carefully considered how each network component might become a potential point of failure or vulnerability during this project, and I created safeguards to mitigate those risks while still meeting the operational needs of the business. Stakeholders can comprehend the risks and the benefits of the suggested safeguards thanks to my ability to evaluate network architecture, apply industry security principles, and clearly explain technical solutions. By combining technical knowledge with structured problem-solving

and diligence, I am confident that I can help JS Corporation maintain a secure, dependable, and scalable network environment as the company continues to grow.

10. References

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